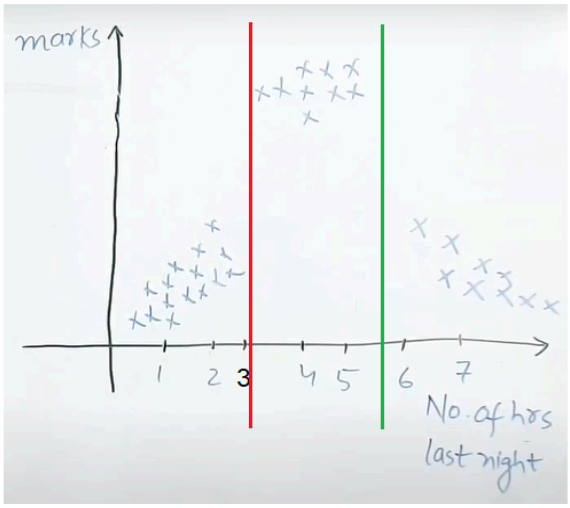
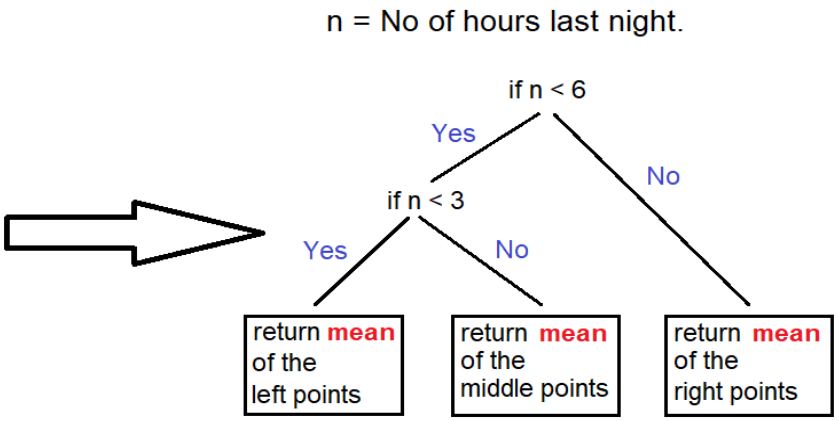
Introduction (Below pictures drawn & explanation learnt from campusx)

When the output column is Numeric (e.g. price, cgpa) instead of categorical, we apply Regression Tree which is a type of Decision Tree.

If the data has a linear relationship, we can easily apply Linear Regression. But if it is Non Linear, we can either apply Polynomial Regression which takes up a huge space as the number of columns increases or Regression Tree.

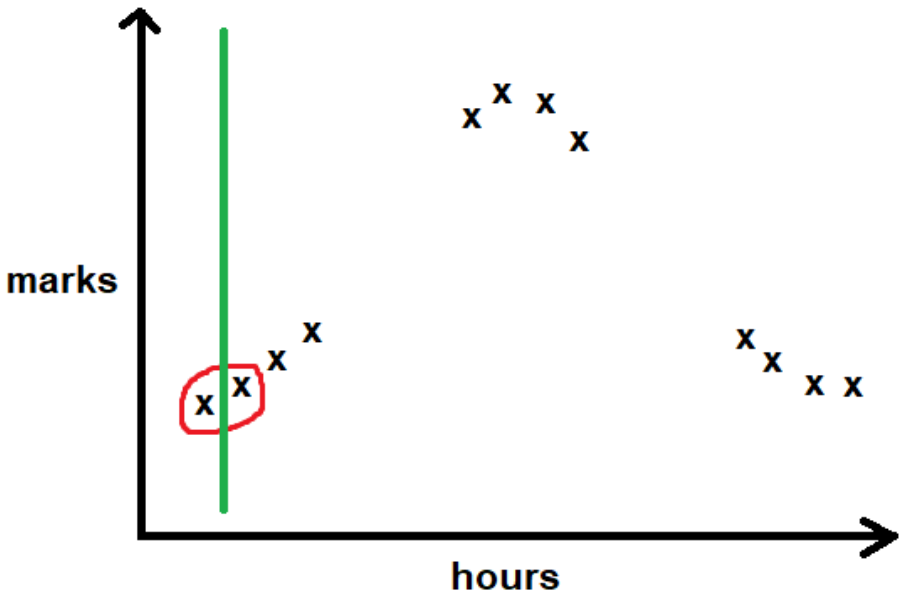




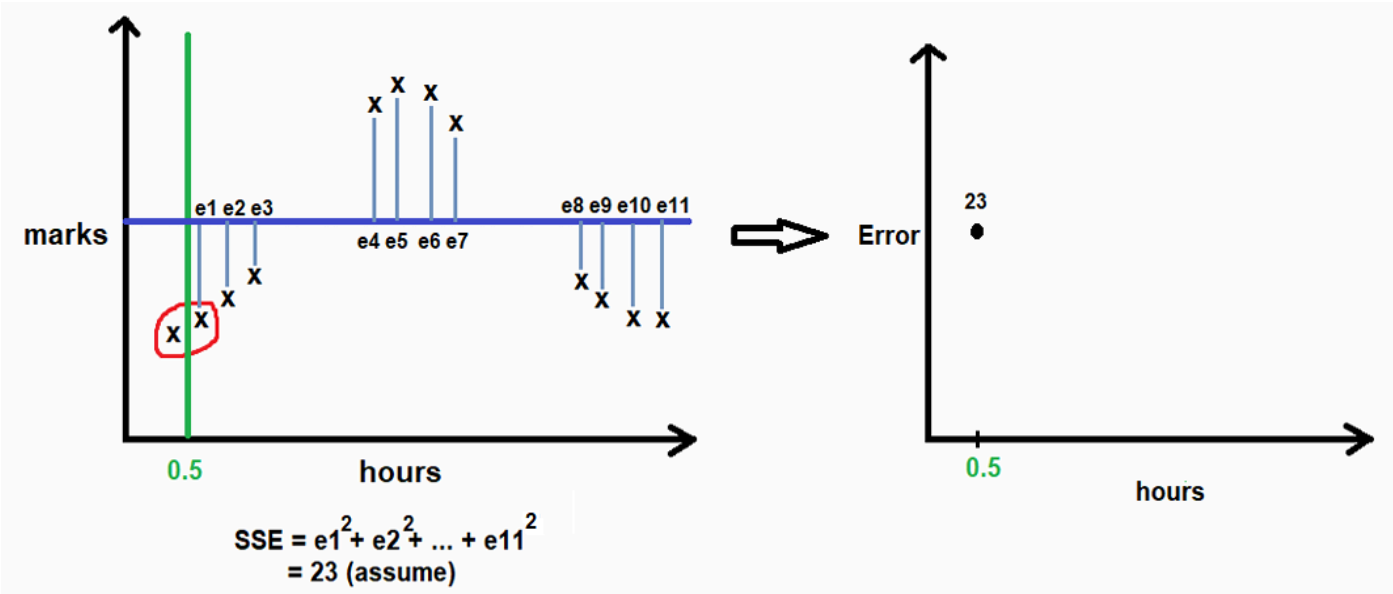
- At the left : We've Non Linear Data. We cut|splitted the data with red and green lines separating the groups. Now if n = 5, we return the **mean** of the middle group of points. That's how Regresion Tree works.
- But the question is how do find the find the **optimal cut|split point** to cut the data?

Optimal cut|split point

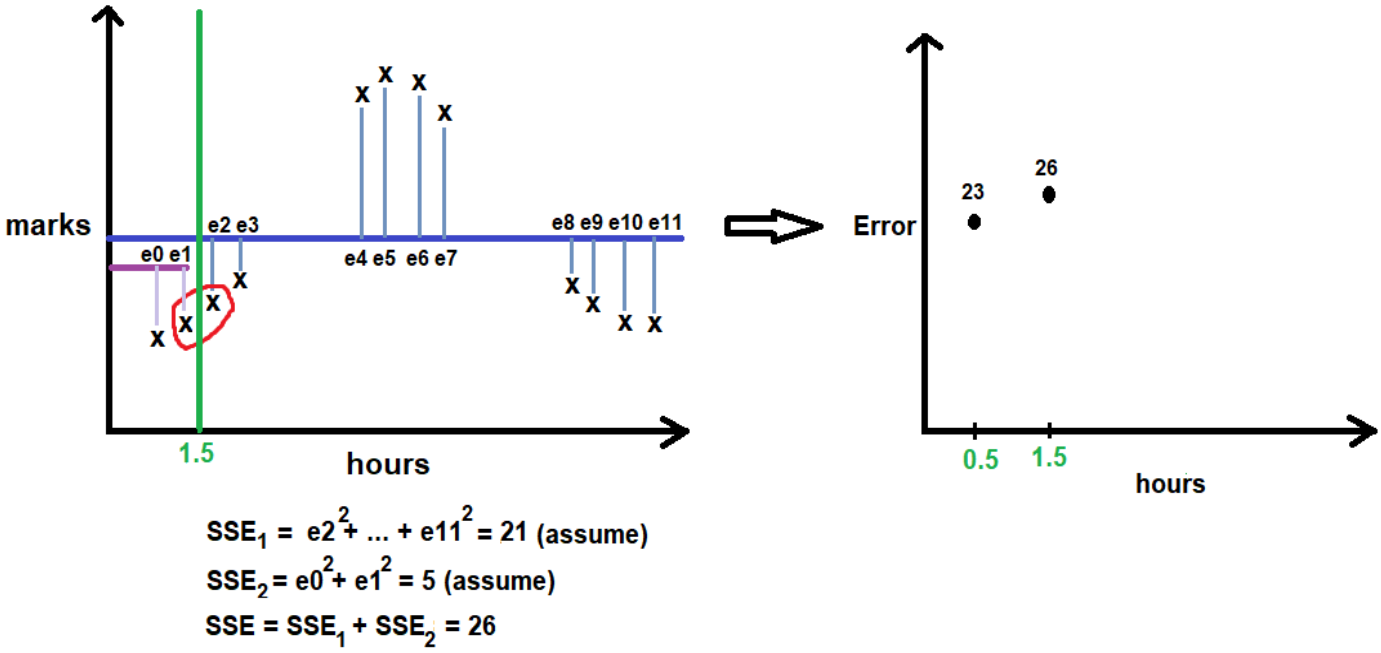
Process for each Split:



Take the first 2 points and separate them with the green line. At the left of the green line, we've only 1 point and at the right of the green line we've more than 1 point. So, for all the right points now we'll calculate SSE (sum of squared error) :

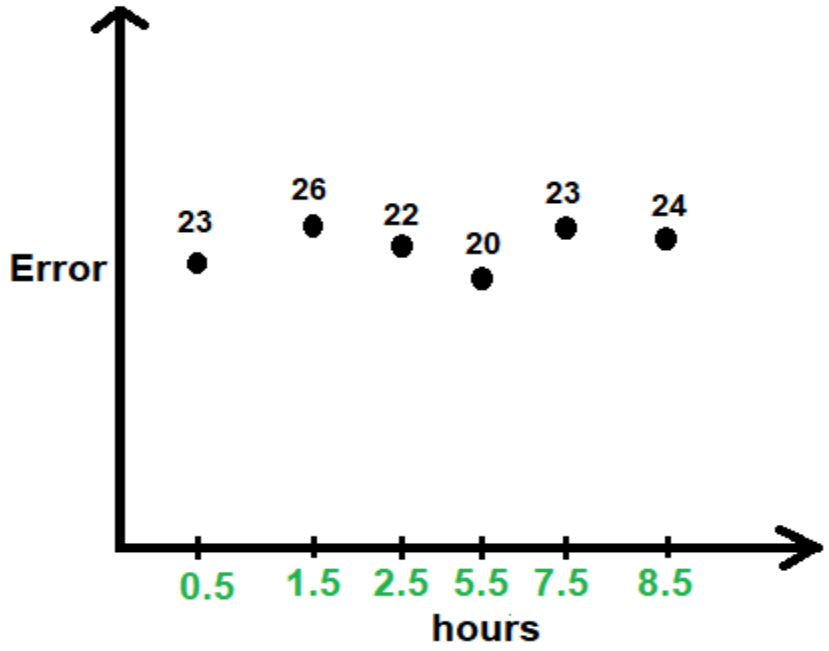


Blue line is the “assumed **mean** marks of all the right points of the green line”.
Now again we’ll take the next 2 points and quantify the SSE for them :

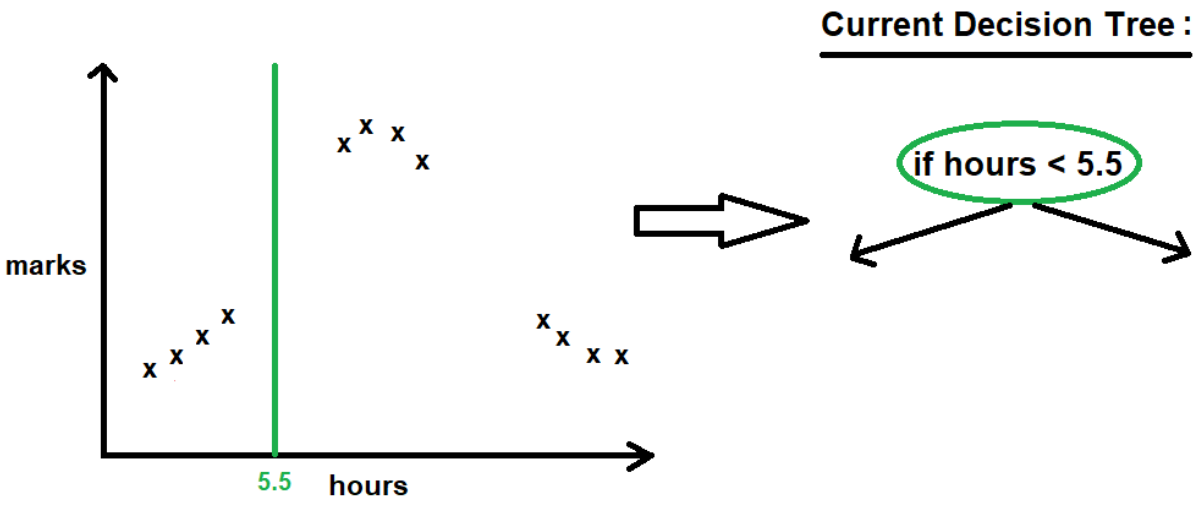


Here there are more than 1 point at the left of the green line and purple line is the **mean** mark of those left 2 points. Blue line is the **mean** mark of all the right points (e2 … e11).

Similarly lets plot all the error for all the next consecutive 2 points :

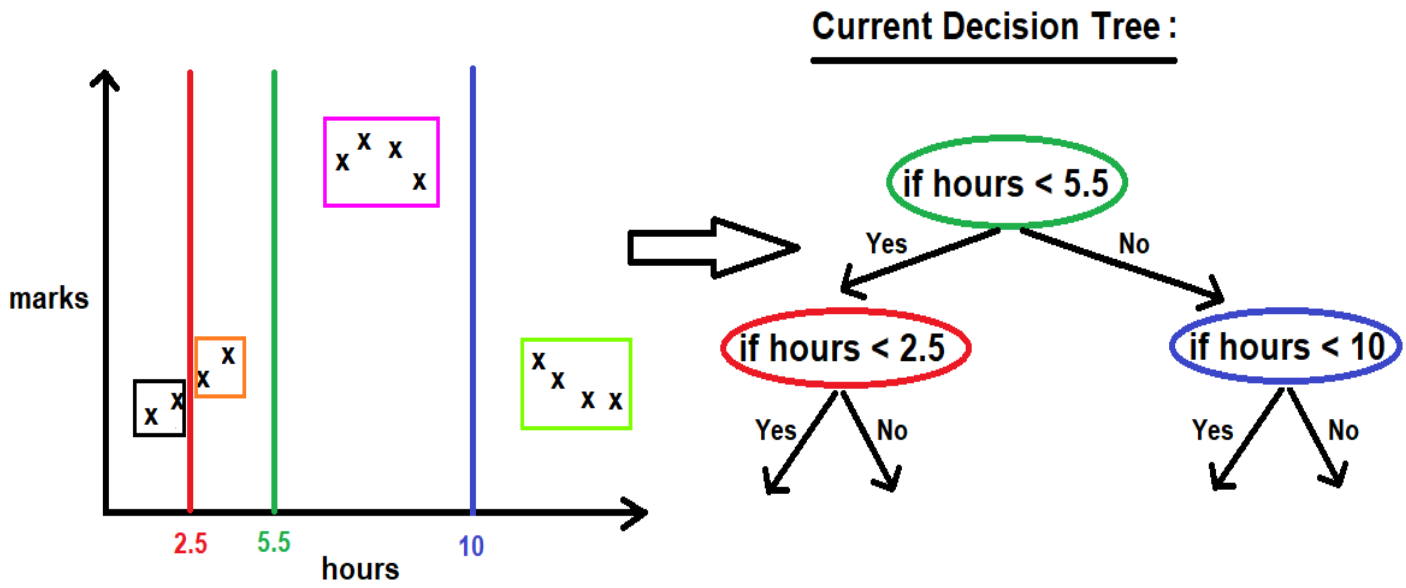


(For 12 points, we'll have 11 Errors but for simplicity purposes I've drawn for the first six)
20 is the minimum Error which belongs to the green line that separates 4th and 5th point. So this is our first split|cut :



Now that we've got our first|root node, we need to find its left and right node(where we'll put condition). The left and right node can be either Decision node or Leaf node.

For the left node's value, we need to find it obviously in the left side of the **Green Splitting Line** and to find the right node's value, we need to find it obviously in the right side of the **Green Splitting Line**. Let's do that :

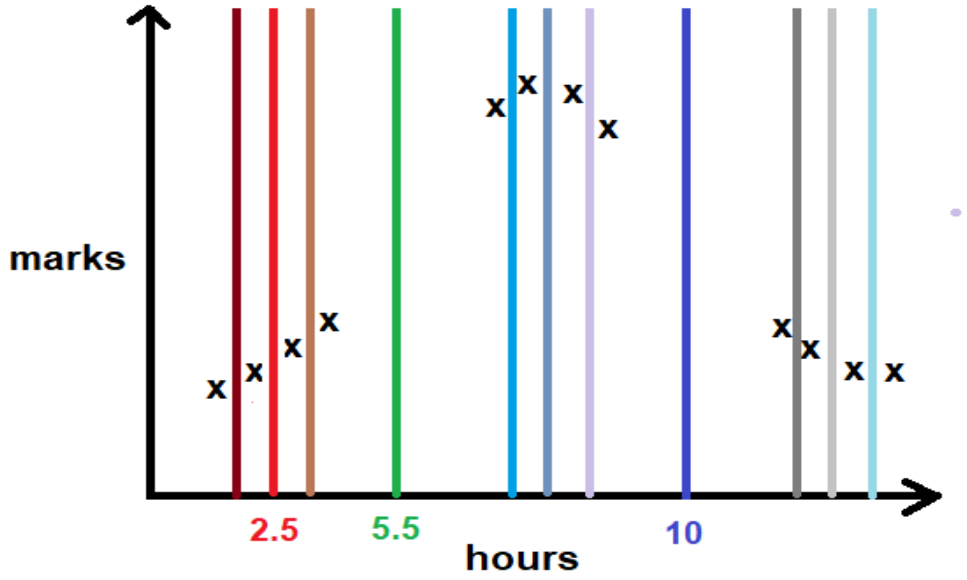


I splitted the left side of the Green line using the Red line and the right side using the Blue Line. Hence we got 2 Decision nodes, red and blue nodes.

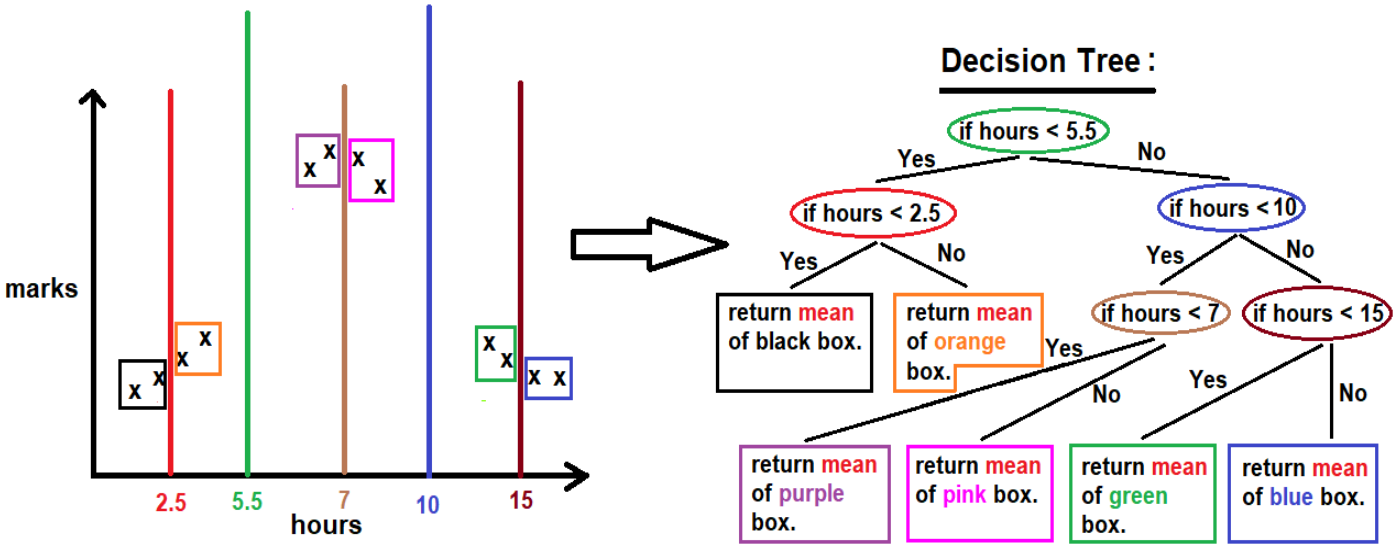
To find the Red Node's (if hours < 2.5) Left and Right node's value, we need to look it up in the Black Box and Orange Box respectively.

To find the Blue Node's (if hours < 10) Left and Right node's value, we need to look it up in the Pink Box and Lime Box respectively.

If we keep splitting like this, we'll have :



So if we keep splitting like this, we'll have a splitting line for each Point. This will perform very well in training data but not testing data, overfitting. And we don't want that. We want to split only if a group i.e. the box has minimum e.g. 3 points :



By default the constant value is around 20 - 25 points|samples|rows campusx said in their video for each group to be eligible to split.

There is a **splitter** parameter in sklearn's DecisionTreeRegressor which is ['best', 'random']. "best" is the way we've done above => Calculating SSE for each two points and then finding the minimum error(to find the corresponding X Axis value) to find the best splitting point. "random" => Instead of the "best" method, just

randomly pick a Point on X Axis and assume it is the optimal Splitting Point. It's used to reduce the Overfitting.

Here we’ve calculated for 1 Input column and 1 Output column. What if we’ve more than 1 Input column? In that case for each input column(X Axis) vs output column(Y Axis), we’ve to find the Minimum Error and then finally find the minimum error of all the Minimum errors. The (Input column, Output column) variation that gives the Minimum Error, parallel to that dimension we’ve to cut|split. This is what campusx said.